

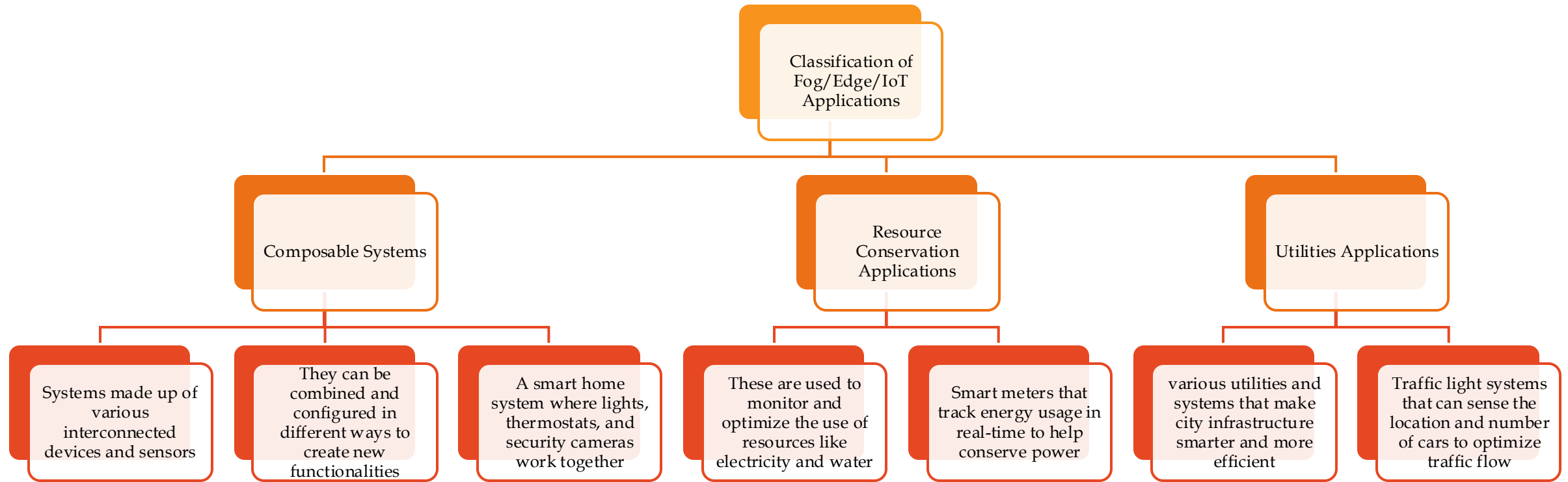
Fundamentals of Fog and Edge Computing

BCSE313L

WINTER 2024 - 2025

Module 7 - Applications of Fog and Edge Computing

Classification of Fog/Edge/IoT Applications



General Data Protection Regulation (GDPR)

- The General Data Protection Regulation (GDPR) is a comprehensive data protection law enacted by the European Union (EU) that went into effect in May 2018

General Data Protection Regulation (GDPR)

Purpose

- Modernize data protection rules to keep up with new technologies
- Strengthens individuals' control over their personal data and reduces administrative burdens

Scope

- GDPR applies not just within the EU but also to organizations outside the EU if they offer services to EU citizens
- It sets a uniform data protection standard across all member states

General Data Protection Regulation (GDPR) – Key Definitions

Personal Data

- Any information related to an identified or identifiable person.

Data Subject

- The individual whose data is being processed.

Controller

- The entity that determines the purposes and means of processing personal data.

Processor

- The entity that processes personal data on behalf of the controller.

Pseudonymization

- Processing personal data so that it can't be attributed to a specific person without additional information.

Limitation

- Restrictions on the purpose, quality, and duration of data storage based on necessity.

General Data Protection Regulation (GDPR) – Consent

- For data processing to be lawful, consent must be given by the data subject for specific purposes.
- Consent should be freely given, specific, informed, and unambiguous.
- Data subjects have the right to withdraw consent at any time.

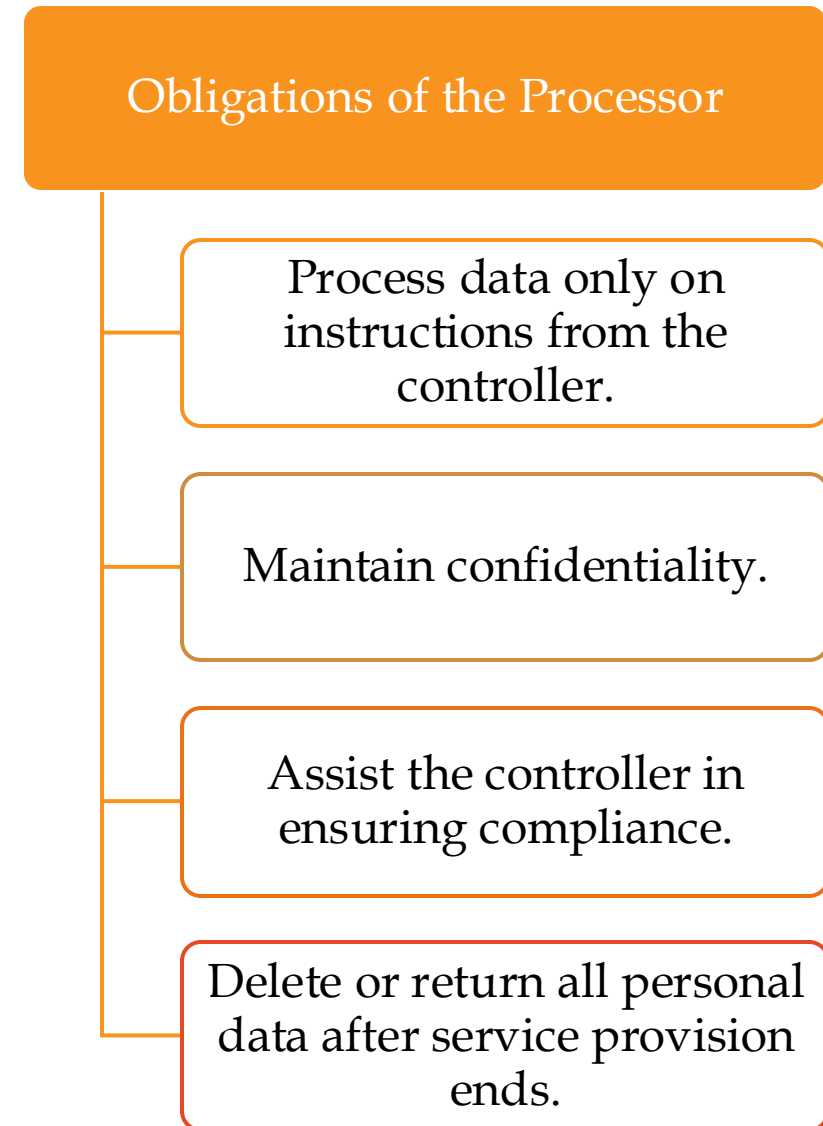
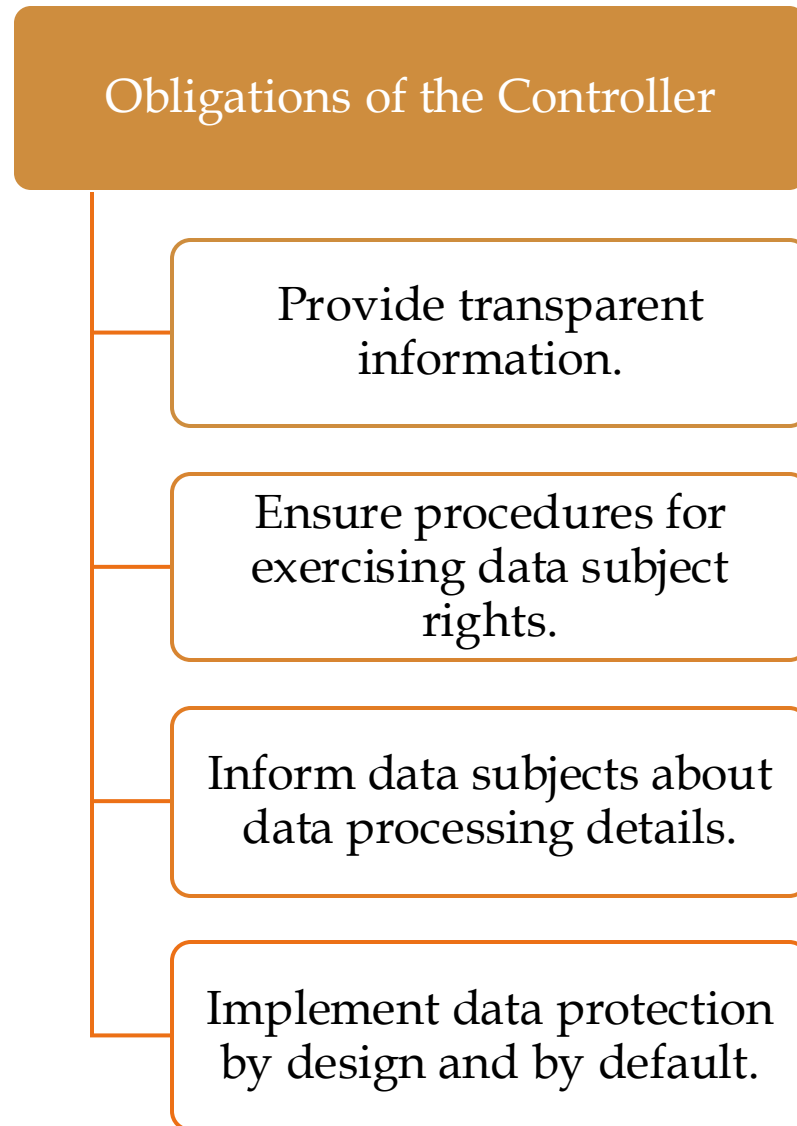
General Data Protection Regulation (GDPR) – Right to be Forgotten

- Individuals have the right to have **their personal data erased** when it's **no longer necessary** or when they **withdraw consent**.
- Controllers must inform third parties to erase links or copies of the data.

General Data Protection Regulation (GDPR) – Data Portability

- Individuals have the right to transfer their data from one electronic system to another without hindrance.

General Data Protection Regulation (GDPR) – Obligations Defined by the GDPR



General Data Protection Regulation (GDPR) – Obligations Defined by the GDPR

- **Record of Processing Operations** : Maintain written/electronic records instead of general notifications to supervisory authorities.
- **Data Breach Notification** : Notify personal data breaches to supervisory authorities and possibly to data subjects.
- **Data Protection Impact Assessment (DPIA)** : Conduct DPIAs for risky processing operations that could significantly affect data subjects' rights and freedoms.
- **Data Protection Officer (DPO)** : Designate a DPO for public sector entities or large enterprises dealing with special categories of data or regular monitoring.

General Data Protection Regulation (GDPR) – Data Transfers Outside the EU

- Personal data can only be transferred to third countries if adequate protection is ensured.
- Conditions for transfers include explicit consent from the data subject, necessity for contract performance, or important public interest reasons.

General Data Protection Regulation (GDPR) – Remedies, Liabilities, and Sanctions

- Data subjects have the **right to judicial remedies** against **controllers or processors**.
- Non-compliance can result in **compensation for damages** and **administrative fines and Penalties**.

General Data Protection Regulation (GDPR) – How it Impacts Cloud & Fog

GDPR Aspect	Definition	Cloud	Fog
Data Minimization & Purpose Limitation	Applications must collect only the minimum necessary data and process it only for specified purposes.	Ensure that data collection is minimized and purpose-specific	It challenging due to distributed nodes
Consent Management	Explicit, informed, and unambiguous consent is required for data processing	Implement robust consent management systems	Obtaining and managing consent across distributed nodes adds complexity
Right to be Forgotten	Users can request deletion of their data	Mechanisms must be in place to erase data across all nodes and backups	Ensuring data deletion from multiple geographically dispersed nodes is crucial
Data Portability	Users have the right to transfer their data.	Systems should support exporting user data in a common format	Aggregating data from various nodes for portability can be complex
Data Protection by Design and Default	Privacy considerations must be integrated into the system design from the outset.	Adopt Privacy by Design principles	Designing nodes to handle data securely and privately

General Data Protection Regulation (GDPR) – How it Impacts Cloud & Fog

GDPR Aspect	Definition	Cloud	Fog
Data Transfers Outside the EU	Personal data can only be transferred outside the EU if adequate protection is ensured.	Ensure compliance with international data transfer rules.	For fog nodes located in different regions, verifying adequacy decisions or implementing safeguards like Binding Corporate Rules (BCRs) is necessary.
Data Breach Notification	Organizations must notify authorities and affected individuals of data breaches within 72 hours	Implement real-time monitoring and alert systems	Detecting and responding to breaches across distributed nodes is essential.
Data Protection Impact Assessments (DPIAs)	DPIAs are required for high-risk processing activities.	Conduct DPIAs to identify and mitigate risks	Assess risks at each node and the overall network.
Appointment of Data Protection Officers (DPOs)	Certain organizations must appoint DPOs.	Large-scale data processors and public sector entities need DPOs.	Coordinating with a DPO across multiple nodes may be necessary.

General Data Protection Regulation (GDPR) – How it Impacts Cloud & Fog

GDPR Aspect	Definition	Cloud	Fog
Record Keeping	Maintain records of processing activities	Document data flows, processing activities, and compliance measures	Tracking data across multiple nodes adds complexity
Legal Roles and Responsibilities	Clear definitions of controllers and processors with respective responsibilities	Define roles clearly, especially in multi-tenant fog environments where multiple entities might be involved.	Define roles clearly, especially in multi-tenant fog environments where multiple entities might be involved.

General Data Protection Regulation (GDPR) – How it Impacts Cloud & Fog

Challenge in Fog-Edge Computing	Description
Distributed Data Processing	Data is processed across multiple nodes, making tracking difficult.
Data Minimization	Collecting only necessary data is hard in distributed environments.
Consent Management	Obtaining and managing user consent across distributed nodes is complex.
Right to Be Forgotten	Deleting data across multiple nodes and caches is operationally challenging.
Data Portability	Aggregating and exporting user data from multiple nodes is difficult.
Data Security	Resource constraints make implementing robust security measures harder.
Data Breach Notification	Detecting and responding to breaches in real-time across distributed nodes is challenging.
Privacy by Design and Default	Embedding privacy into the design of a dynamic and heterogeneous system is difficult.

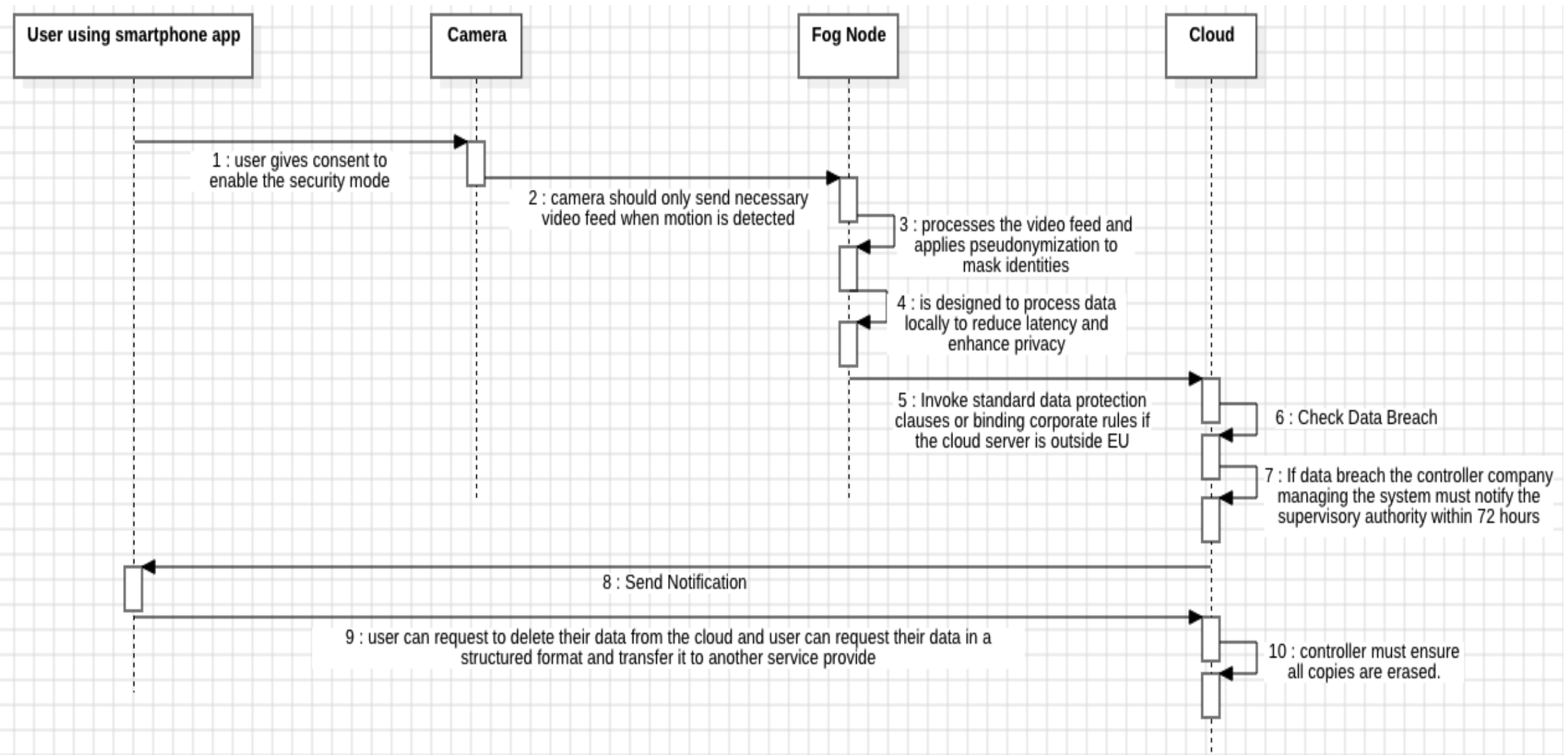
General Data Protection Regulation (GDPR) – How it Impacts Cloud & Fog

Challenge in Fog-Edge Computing	Description
Data Transfers Outside the EU	Nodes may be in regions without adequate data protection, complicating compliance.
Resource Constraints	Limited processing power, memory, and bandwidth hinder implementation of GDPR-compliant mechanisms.
Dynamic and Heterogeneous Environment	Nodes frequently join or leave the network, making consistent compliance hard.
Lack of Centralized Control	Decentralized architecture complicates enforcement of GDPR policies.
Real-Time Processing Requirements	Balancing GDPR compliance with low-latency requirements for real-time applications is challenging.

General Data Protection Regulation (GDPR) – Smart Home Security System

- Imagine a smart home security system that uses IoT devices (like cameras and sensors), processes data in a fog node (a local server in the house), and stores some data in the cloud for remote access and long-term storage.

General Data Protection Regulation (GDPR) – Smart Home Security System



Testing Perspectives of Fog

- The various approaches, methodologies, and considerations involved in ensuring the reliability, security, performance, and usability of applications that operate within a fog computing environment
- Distributed and decentralized nature introduces unique challenges that must be addressed through rigorous testing

Testing Perspectives of Fog – Testing Objectives

- Ensure that the fog-based applications meet specific goals like:
 - **Low Latency** : Ensuring real-time processing and responsiveness.
 - **Interoperability** : Verifying seamless communication between fog nodes, IoT devices, and cloud systems.
 - **Security and Privacy** : Protecting sensitive data from breaches and ensuring compliance with regulations like GDPR.
 - **Reliability** : Guaranteeing uninterrupted service even with intermittent connectivity or node failures.
 - **Scalability** : Supporting massive machine-to-machine communications and large-scale IoT deployments.
 - **Energy Efficiency** : Optimizing energy consumption for sustainable operation.

Testing Perspectives of Fog – Testing Types



Testing Perspectives of Fog – Testing Types

Functional Testing

- Validates whether each function of the application works as intended

Performance Testing

- Assesses how well the system performs under expected workloads

Security Testing

- Ensures the application is secure from cyber threats and vulnerabilities

Interoperability Testing

- Confirms that different components of the fog ecosystem (e.g., IoT devices, fog nodes, cloud servers) can communicate effectively

Testing Perspectives of Fog – Testing Types

Usability Testing

- Evaluates how user-friendly the application is for end-users

Energy Efficiency Testing

- Measures the energy consumption of fog nodes and IoT devices.

Reliability Testing

- Ensures the system remains stable under stress conditions or device failures.

Conformance Testing

- Checks compliance with industry standards and regulations.

Testing Perspectives of Fog – Testing Challenges

Distributed Architecture

- Data is processed across multiple nodes, making it harder to track and test.

Resource Constraints

- Fog nodes often have limited processing power, memory, and bandwidth, complicating the implementation of robust testing tools.

Dynamic Environment

- Nodes frequently join or leave the network, requiring adaptive testing strategies.

Heterogeneity

- Devices and platforms in fog environments are diverse, necessitating extensive compatibility testing.

Privacy Concerns

- Ensuring privacy in geographically distributed systems where location tracking may occur.

Real-Time Processing

- Balancing low-latency requirements with thorough testing.

Testing Perspectives of Fog – Testing Approaches

➤ **Simulation-Based Testing**

- Uses virtualized environments to simulate fog nodes and IoT devices.
- Example: NREL's smart home test bed simulates power distribution grids.

➤ **Test Beds**

- Physical or virtual setups designed to replicate real-world scenarios for testing.
- Example: ETSI test beds for smart transport, equipped with traffic lights, IoT sensors, and cameras.

Testing Perspectives of Fog – Testing Approaches

➤ Automated Testing

- Leverages tools like Selenium, Appium, and JMeter to automate repetitive testing tasks.
- Example: Automating regression testing for healthcare workflows.

➤ Context-Aware Testing

- Focuses on testing applications in specific contexts (e.g., user location, device type).
- Example: Testing smart glasses for short battery life and usability in different scenarios.

Testing Perspectives of Fog – Testing Approaches

➤ Security-Specific Testing

- Includes intrusion detection/prevention systems, sensor spoofing tests, and secure software updates.
- Example: Dedicated cybersecurity labs for smart transport systems.

Testing Perspectives of Fog – Testing Case Studies

➤ **Smart Homes**

- Objective : Reduce energy consumption and ensure data protection.
- Approach : Combining powered hardware and software simulations (e.g., NREL test bed).
- Outcome : Certified devices with enhanced security and interoperability.

➤ **Smart Health**

- Objective : Ensure secure communication between doctors and patients.
- Approach : Adhering to HIPAA regulations and performing interoperability testing.
- Outcome : Reliable healthcare workflows and compliant mobile apps.

➤ **Smart Transport**

- Objective : Develop collision avoidance algorithms and test autonomous vehicles.
- Approach : Using test beds like UMTRI and US DOT Connected Vehicle Test Bed.
- Outcome : Improved vehicle safety and reduced traffic congestion.

Testing Perspectives of Fog – Testing Case Studies

Domain	Testing Types
Smart Homes	Energy efficiency, reliability, functionality, interoperability, security, performance, usability
Smart Health	Functional, interoperability, enterprise workflow, performance, usability, security, compliance
Smart Transport	Collision avoidance, autonomous vehicle, performance, usability, pollution control, interoperability, privacy, energy efficiency, reliability, security

Case Study – Intelligent Smart Traffic System

➤ Objective

- The Intelligent Traffic Lights Management (ITLM) system aims to deploy intelligent traffic lights (ITLs) equipped with **sensing capabilities at intersections**.

➤ Key QoS Attributes

- **Accident Prevention** : ITLs must **trigger stop or slow-down signals to prevent collisions in real time**.
- **Ensuring Vehicle Mobility** : ITLs need **efficient software programming interfaces** to learn **fleet dynamics** and **maintain steady traffic flow**.
- **Reliability** : **Historical datasets** generated by ITLM systems are **analyzed using big data analytics (BDA) tools** to evaluate and **enhance system reliability**.

Case Study – Intelligent Smart Traffic System

➤ Computing Requirements of ITLM Systems

Attributes	Description
Mobility	Tight mobility constraints for commuters and ITLs (e.g., regular red-green pulse cycles).
Geo-Distribution	Wide (across regions) and dense (at intersections and ramp accesses).
Low/Predictable Latency	Tight latency requirements within the scope of intersections.
Fog-Cloud Interplay	Data handled at different time scales (e.g., real-time sensor/vehicle data vs. aggregated traffic information).
Multi-Agency Orchestration	Agencies running the system must coordinate control policies in real time.
Consistency	Achieving consistency between data collection points is crucial for implementing efficient traffic policies.

Case Study – Intelligent Smart Traffic System

➤ Fog Model for ITLM

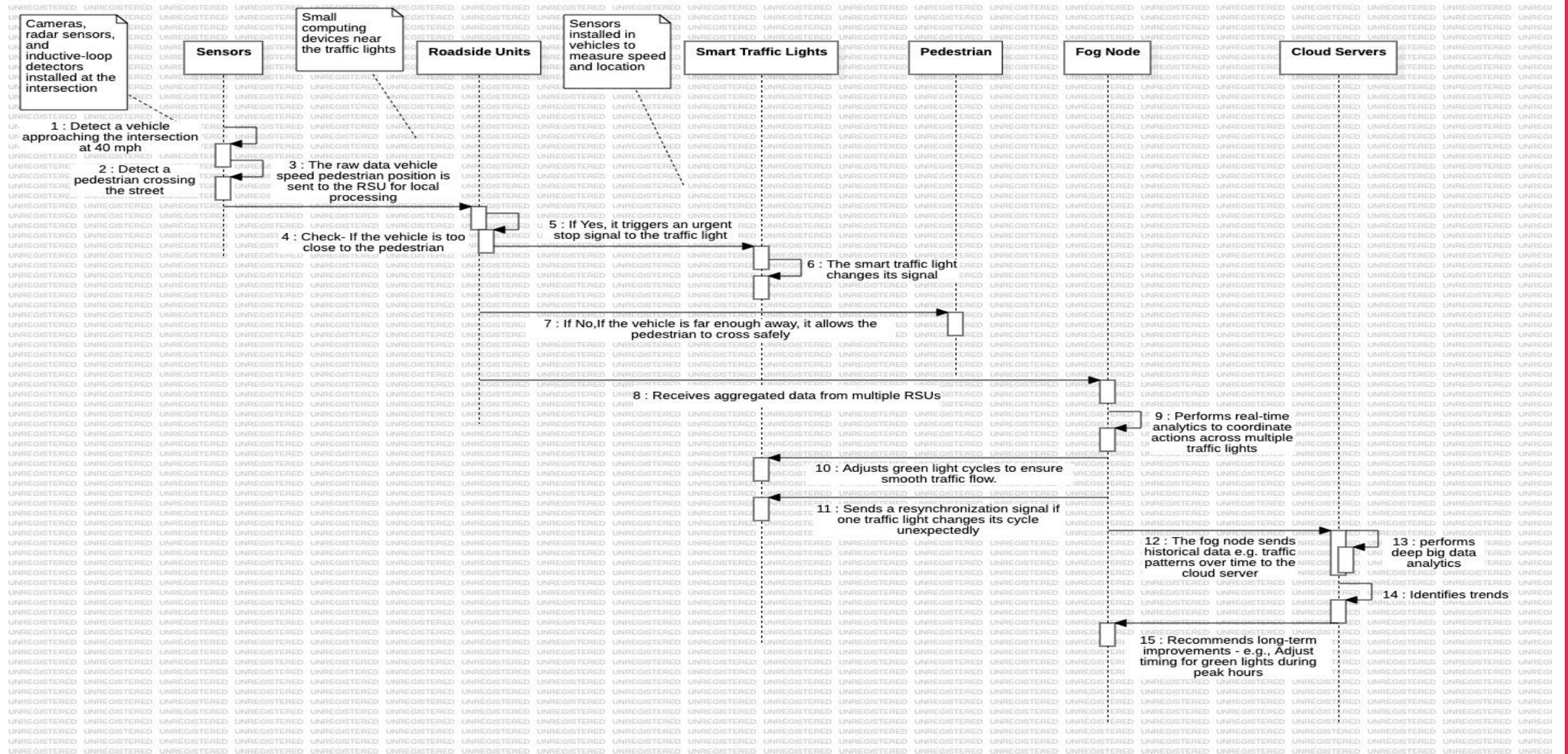
Attributes	Description
Modular Compute and Storage Devices	▪ The fog model provides common interfaces and programming environments for ITL networking infrastructures. ▪ It supports varying form factors and encasings while ensuring consistency between aggregator points.
Strict Service Guarantees	ITL fog nodes provide strict service guarantees for mission-critical systems like ITLM, unlike softer guarantees (e.g., infotainment).
Multi-Tenancy	ITL networks often extend beyond the domains of a single controlling authority, requiring orchestration of consistent policies across multiple agencies

Case Study – Intelligent Smart Traffic System

➤Orchestration Framework for ITLM

Attributes	Description
Cloud-Fog Dispatch Middleware (CFDM)	▪ Acts as an orchestration platform to handle critical software components across the system. ▪ Deployed over a wide geographical area.
Decision-Making Modules (DMM)	▪ Create control policies and push them to individual ITLs. ▪ Can be implemented in centralized, distributed, or hierarchical ways.
Federated Message Bus Protocols	• Pass data from traffic lights to DMM nodes. • Push policies from DMM nodes to ITLs. • Exchange information between ITLs.

Case Study – Intelligent Smart Traffic System



Case Study – Intelligent Smart Traffic System

